Ceph

THE FUTURE OF STORAGE

ghostwydwu （武勇栋）



Ceph

1. what is ceph
   Ceph is a unified, distributed storage system designed for excellent performance, reliability and scalability

2.2. Key Features
- **Unified storage**: object storage, block storage, POSIX-compliant file-system interfaces.
- **High scalability**: commodity x86 servers, cluster size from tens to over or thousand nodes; capacity from TB to PB scale.
- **High reliability**: no single point of failure; multi-replica data; automatic self-management and self-healing.
- **High performance**: well-balanced data distribution, high parallelism. Obje and block-storage paths do **not** require dedicated metadata servers.2. Ke Features
- **Unified storage**: object storage, block storage, POSIX-compliant file-system interfaces.

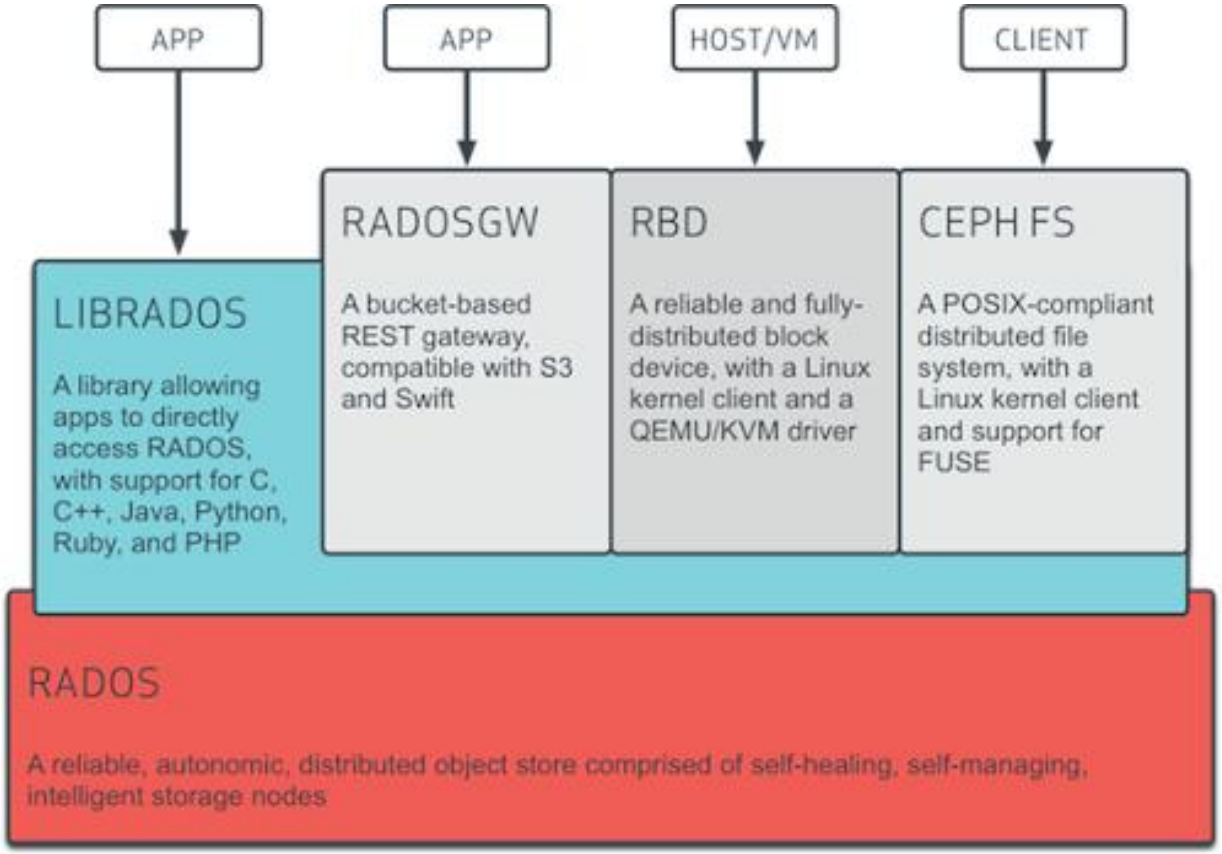
腾讯网 QQ.com

Ceph

- **High scalability**: commodity x86 servers, cluster size from tens to over o
thousand nodes; capacity from TB to PB scale.
- **High reliability**: no single point of failure; multi-replica data; automatic
self-management and self-healing.
- **High performance**: well-balanced data distribution, high parallelism. Obj
and block-storage paths do **not** require dedicated metadata servers.

腾讯网 QQ.com

Ceph Architecture Overview



腾讯网 QQ.com

Ceph Core Roles

- OSD(Object Storage Device): Store actual user data. OSD nodes exchange heartbeats; report peer failures to Monitor.

- Monitor： Maintain global cluster view (OSD Map). Accept status reports from OSDs. Run Paxos-based quorum protocol for consensus.

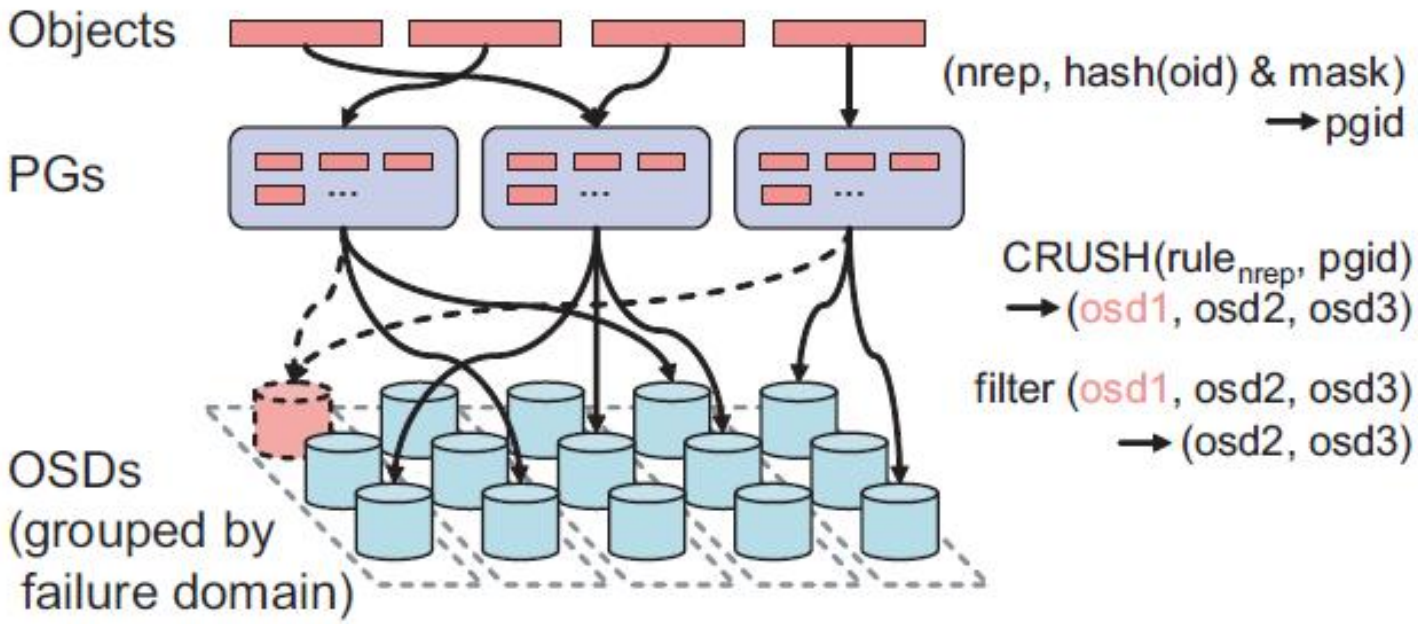
- MDS (Metadata Server): Manage metadata exclusively for CephFS.

- PG(Placement Group): Logical grouping of objects mapped onto a set of OSDs (replica count typically 2 or 3).

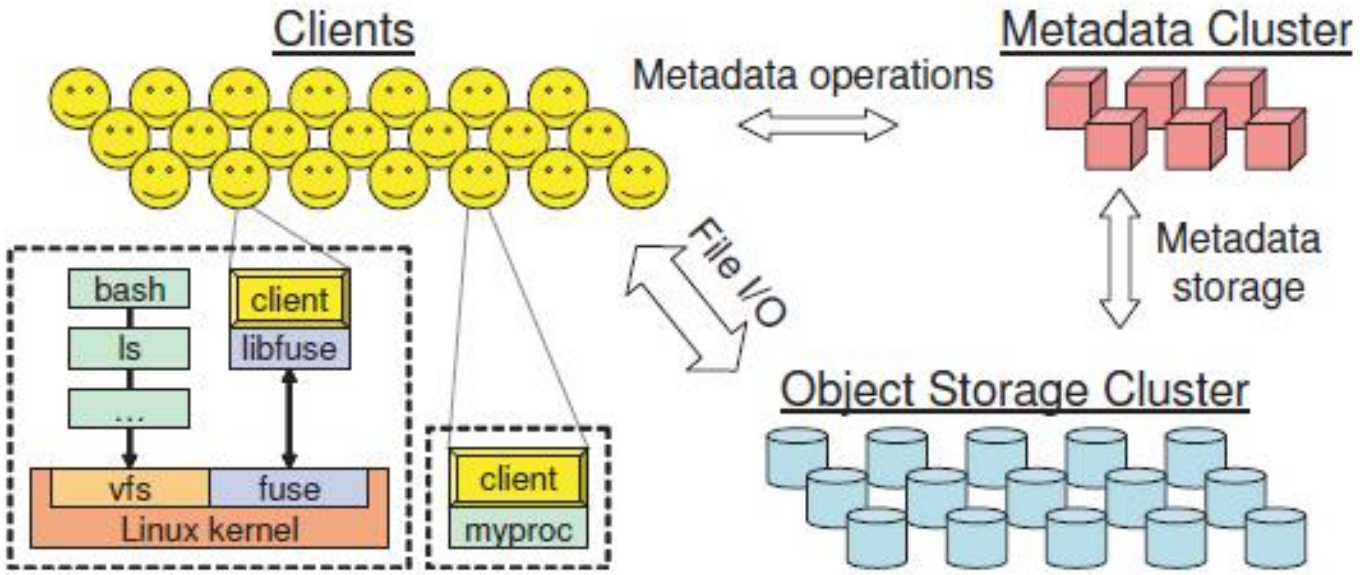
- Pool: Top-level logical partition; collection of many PGs, defines replica count and CRUSH rules.

腾讯网 QQ.com

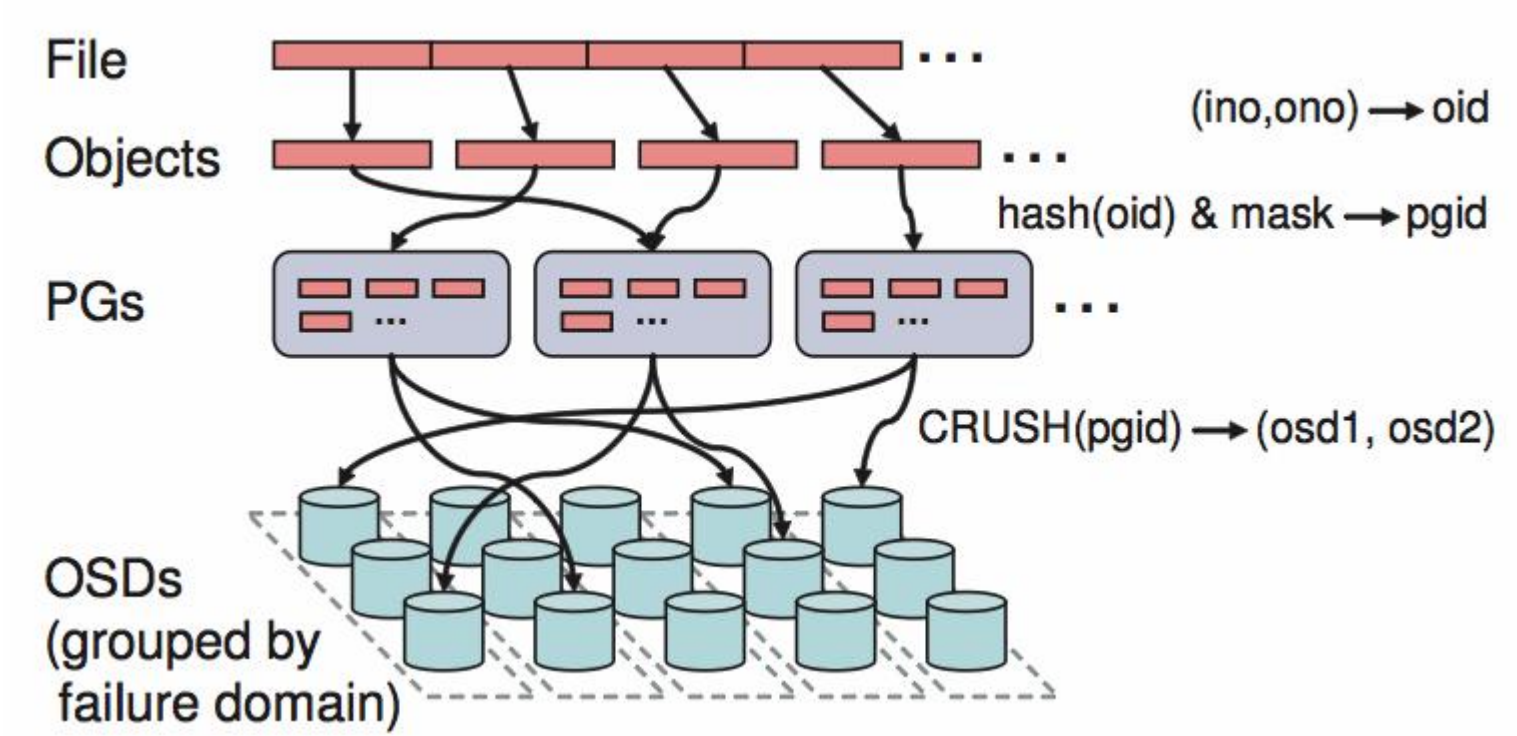
Ceph Object Storage Architecture



Ceph Block Storage Architecture



Ceph File System Architecture



Ceph Data Storage Layers

The logical division in Ceph storage is the **pool**, which is analogous to the concept of a large table.

Pool attributes include: the replica count of objects, the number of Placement Groups (PGs), and the CRUSH RuleSet in use.

Data mapping flow: `(Pool, Object) → (Pool, PG) → OSD set → Disk`

腾讯网 QQ.com

Ceph Data Storage Layers

2. Factors affecting the mapping from **(Pool, PG)** to **OSD set**
- CRUSH algorithm: A pseudo-random algorithm
- OSD MAP: Contains the status of all current pools and all OSDs
- CRUSH MAP: Describes the hierarchical topology of disks, servers and racks in the cluster
- CRUSH Rules: Data placement policies, including constraints such as which racks, data-centers and hosts a pool is allowed to use.

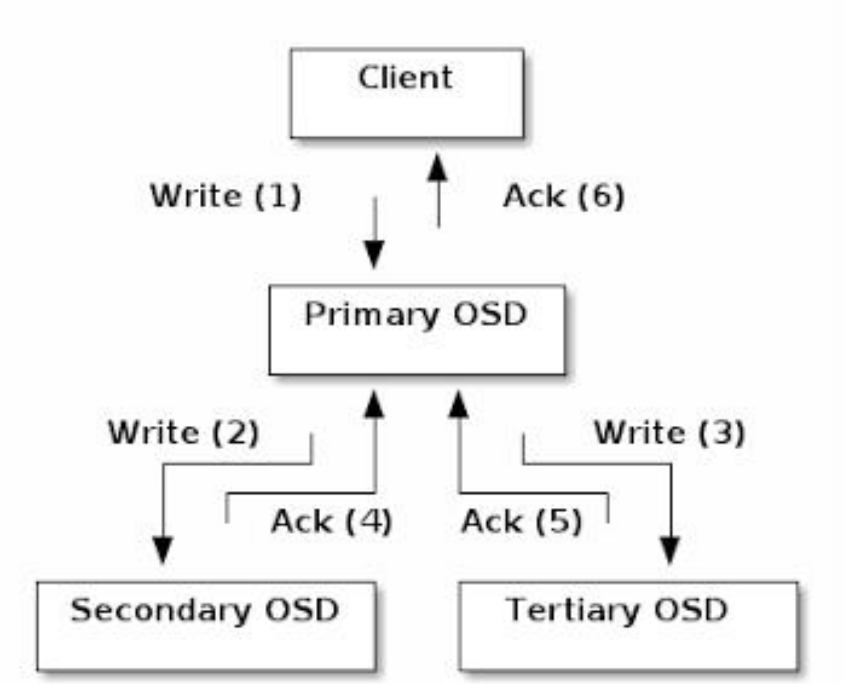
3. data access Steps:
1. Client fetches `OSD Map`, `CRUSH Map`, `CRUSH Rules` from Monitor cluster.
2. Compute PG-id by hashing object OID and mask against pool parameters.
3. Execute **CRUSH** algorithm to deterministically map PG-id to an ordered list of OSD peers (primary + replicas).
4. Client communicates directly with primary OSD, no central routing proxy.

腾讯网 QQ.com

Ceph Write Consistency Model (Primary-Replica)

Strong Consistency

1. Client sends write request to primary OSD.
2. Primary forwards write operation to all replica OSD peers.
3. Each replica persists data and replies ACK to primary.
4. Primary only acknowledges success back to client **after all replicas acknowledge completion**.。



腾讯网 QQ.com

Ceph Route Algorithm--CRUSH

1. Crush概念

CRUSH (Controlled Replication Under Scalable Hashing), a pseudo-random data distribution function that efficiently maps each PG to an ordered list of OSDs upon which to store object replicas.

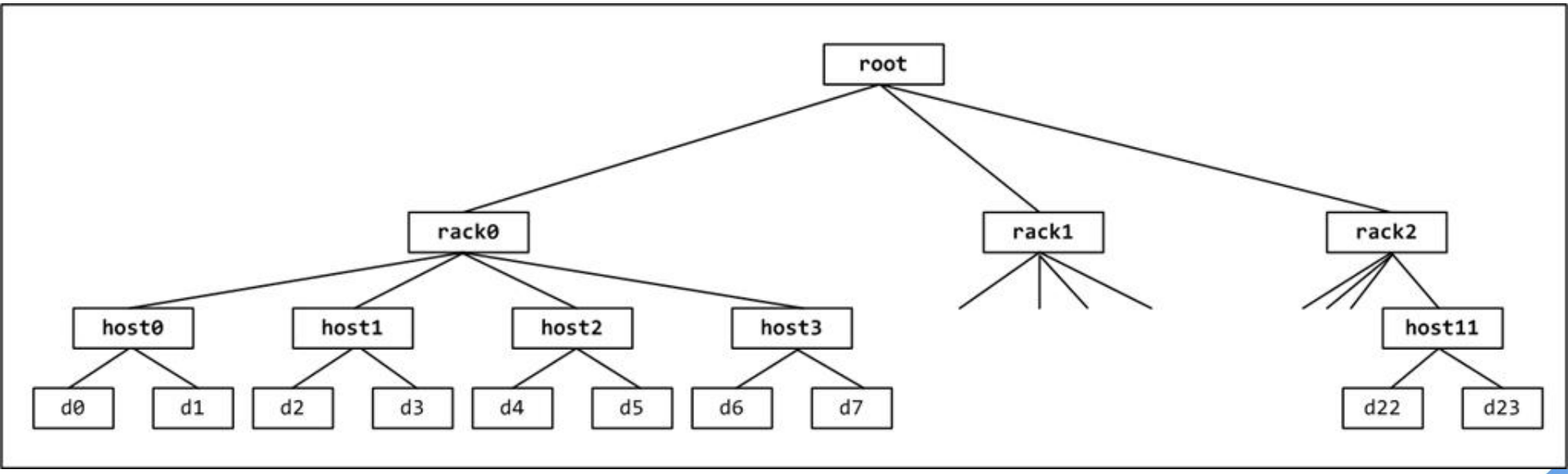
2. Advantage

- Calculation-based placement instead of static routing tables.
- Client holds minimal metadata locally.
- Flexible placement policies: restrict data distribution by datacenter, rack, host, media type.
- Minimize data migration during cluster scaling (add / remove OSD).

腾讯网 QQ.com

CRUSH

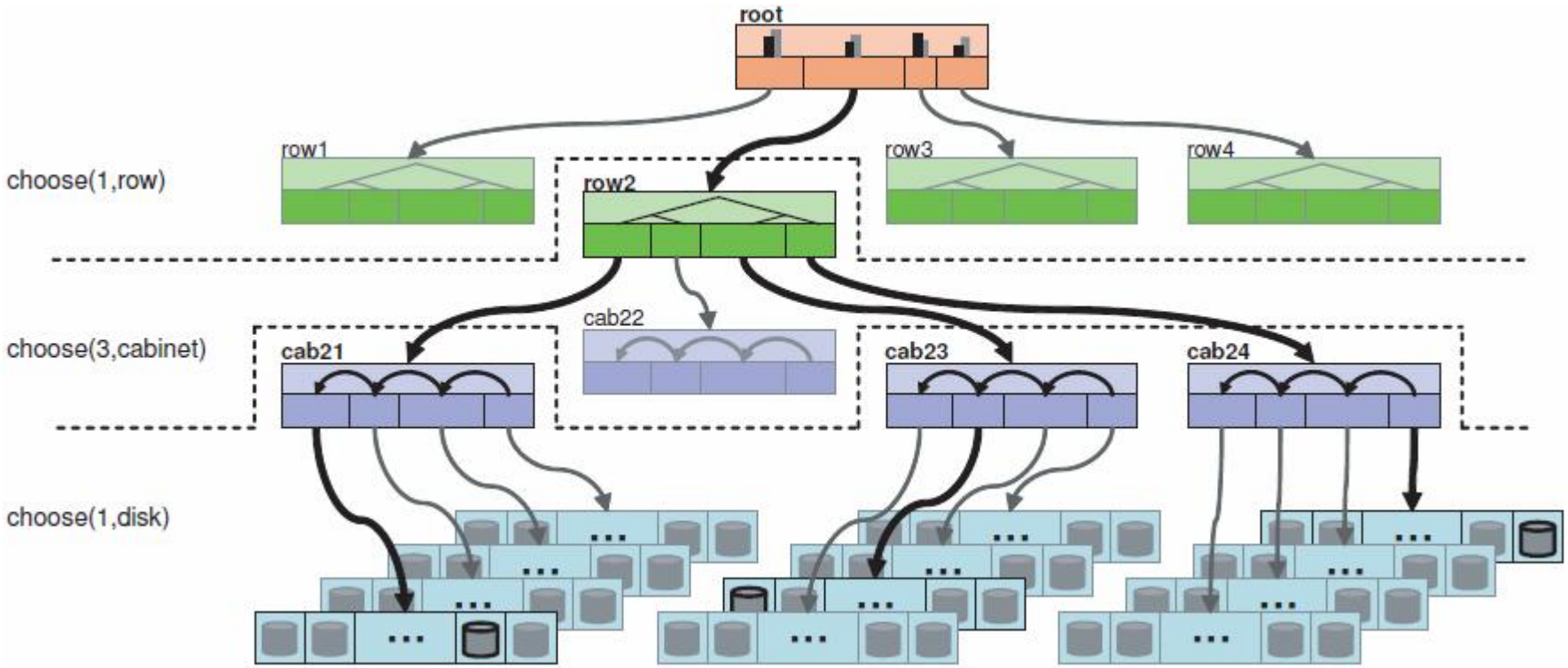
3. CRUSH Map Structure
Diagram: 3 racks (rack0), 4 hosts per rack, 2 devices per host.



腾讯网 QQ.com

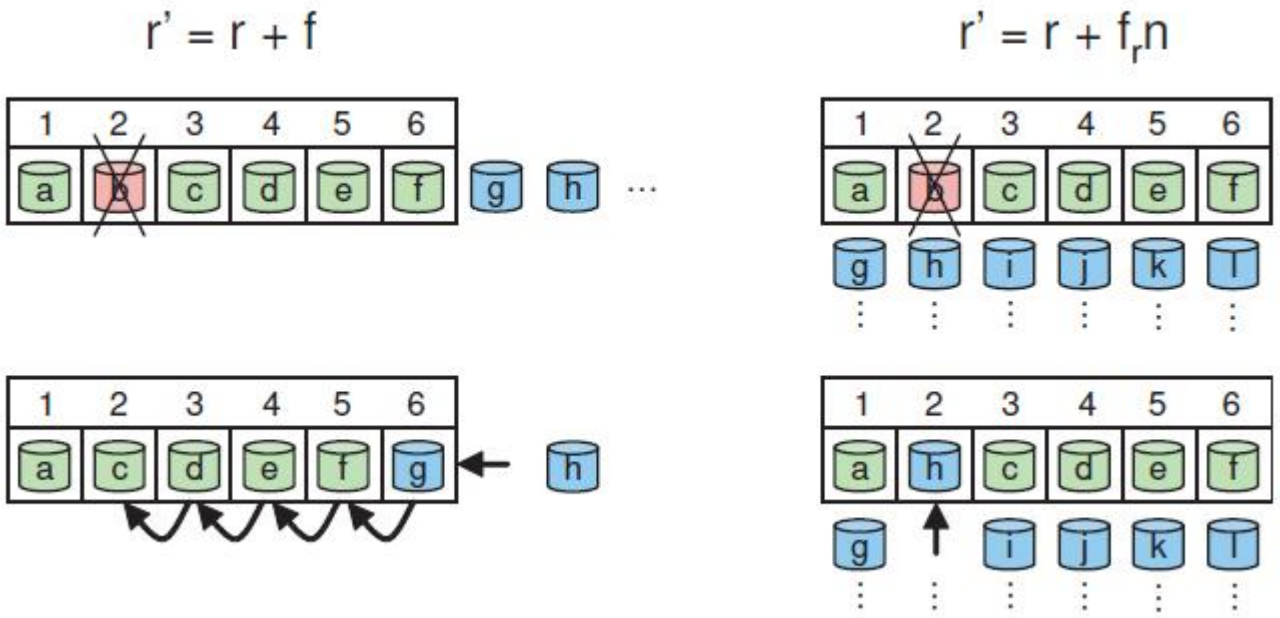
CRUSH

4. OSD Lookup



CRUSH

5. OSD Selection



Ceph Fault Tolerence

Typical failure events: network partition, power loss, server crash, disk hardware failure.

1. Fault-detection**: OSD-to-OSD heartbeat. When peer OSD unresponsive, report failure to Monitor.
2. Monitor marks failing OSD as `down`. After timeout period, mark as `out` and update OSD-Map.
3. If primary OSD fails: one replica is promoted as new primary. PG enters `active+degraded` state.
4. **Peering phase**: new primary collects authoritative PG operation history from surviving peers, makes replicas converge to consistent state.
5. After peering completes: PG enters `active+recovering`. Primary migrates missing objects to bring replica count back to desired value.

腾讯网 QQ.com

Ceph Fault Tolerence

3. Fault Recovery
- When an OSD in the OSD set is marked as down — if the failed OSD is the primary, one replica is promoted as the new primary to handle all object read/write requests — the PG enters the `active+degraded` state.
- If the downed host remains unreachable after M seconds, it is marked as `out`. Ceph then recomputes the PG-to-OSD-set mapping via the CRUSH algorithm.
- The new primary first collects PG operation logs from the surviving OSDs in the original set to derive the **Authoritative History** — a complete, totally ordered sequence of operations. It then reaches consensus with all other replicas on this Authoritative History, i.e. all replicas converge to an identical state for every object contained in the PG. This synchronization process is known as **Peering**.
- After Peering completes, the PG transitions to the `active+recovering` state. The primary migrates and synchronizes all degraded objects to every replica, until the replica count for each object meets the configured value N.

腾讯网 QQ.com

Ceph VS TSSD VS CBS

	TSSDV3	CBS3.0	Ceph
Central Routing Node	YES	YES	NO
Fault Detection & Recovery	external detection	external intervention required	Fully automatic detection & self-recovery
Data Balance	external tools	external tools	Built-in automatic balancing
Access Proxy	YES	YES	YES
Strong consistency	YES	YES	YES
Theoretical maximum capacity	2P（100万*2G）	2P（100万*2G）	Practically unlimited
Data Model	key-value	key-vale	key-value，file system，block storage

腾讯网 QQ.com

Ceph VS TSSD VS CBS

1. Our Advantage
- Core-system code is self-designed and implemented, with strong controllability of production systems.
- Closely tailored to in-house business; optimized for business workloads and hardware models for improved business experience.
- Rich operational experience and high controllability of the full storage platform.

2 Lessons learned from Ceph
1. Study CephFS implementation patterns to deepen understanding of cloud-native storage.
2. Evaluate Ceph fault-detection mechanisms for potential adoption in self-built storage systems.
3. Understand snapshot design patterns; combine with snapshot capability of CBS.
4. Broaden knowledge of OpenStack, Swift, Amazon S3, container-ecosystem storage.

腾讯网 QQ.com